

W4000 Multi-parameter Sensor

User Manual



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Introduction

Thanks for choosing Yantai Winmore instrument !

Please read this manual carefully before using this product, and keep this manual in safe place for future reference.

Please follow the instructions and procedures stated in this manual.

To ensure after sales warranty coverage, please follow the user instructions and maintenance procedures stated in this manual. Any damage and lost caused by improper use of this product will not be covered by factory warranty. Please keep all documents, and if you have any questions, please do not hesitate to contact Yantai Winmore's customer services.

Remove the instrument from package material and examine it to make sure that there is no damage occurred during shipment. If there is any damage, please contact Yantai Winmore Customer Service immediately. Save all materials until you are sure that the instrument functions properly. Any damage or defective items must be returned in their original packaging material.

W4000 Multi-parameter Sensor

Overview

Yantai Winmore has launched the latest online multi-parameter (all-in-one) water quality sensor, integrated design, reliable and easy to use. Up to 6 digital sensors from Winmore can be connected at the same time, fluorescence dissolved oxygen sensor, four-electrode conductivity sensor, turbidity sensor, digital pH electrode, blue Chlorella and Chlorophyll sensor are optional. RS485 output, with automatic cleaning device, greatly reducing maintenance costs, each sensor is equipped with a quick-release waterproof connector. Compared with traditional shore-type instruments, the installation is more convenient, and it can reflect the real data of the scene more intuitively.



Features

- Digital sensor, RS485 output, supporting MODBUS
- All calibration data stored in sensors, each sensor has a waterproof joint, easy to plug and replace.
- Equipped with automatic cleaning device, it can effectively remove the surface contamination of the sensor and prevent the growth of microorganisms, more accurate and less maintenance.
- Fluorescence dissolved oxygen sensor, four-electrode conductivity sensor, turbidity sensor, digital pH electrode, blue Chlorella and Chlorophyll sensor are optional, suitable for long-term online monitoring.
- Integrated design, you can connect 6 probes at the same time and measure 7 parameters.

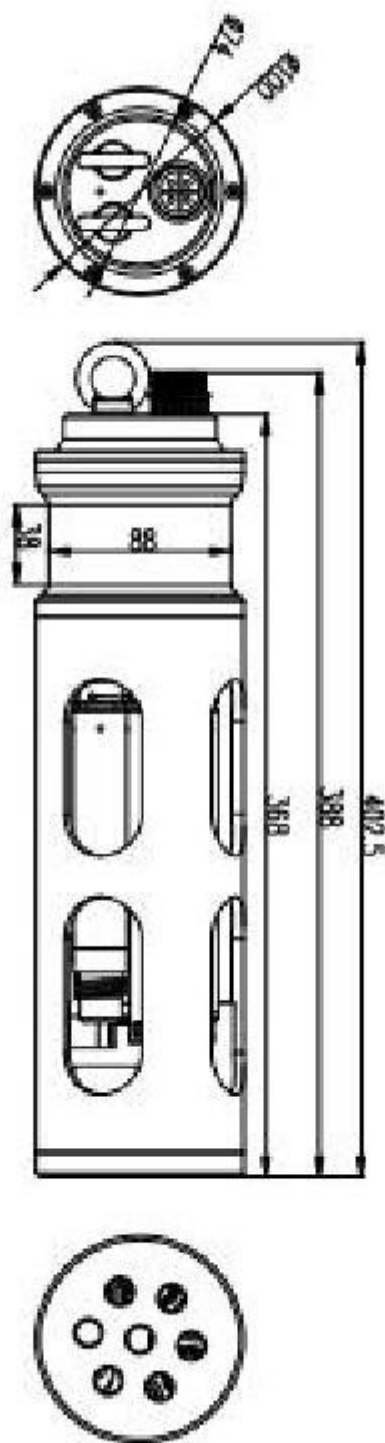


Ordering Information

Product Name	Model	Description
Online multi-parameter (All-in-One) water quality	W4000	Including fluorescence dissolved oxygen sensor, four-electrode conductivity

sensor		sensor, turbidity sensor, digital pH electrode, temperature sensors, and self-cleaning brush. 12V power supply. IP68 waterproof. One channel RS485 output.
Online multi-sensor fluorescence dissolved oxygen sensor	W504-B	Test range is 0~20mg/L or 0~200% saturation, RS485 port, waterproofing connector, IP68.
Online multi-sensor four-electrode conductivity sensor	W520-B	Test range: 1 μ s/cm~200ms/cm, accuracy: <1%, sensor material: nickel, sheath material: PEEK, operating temperature: :0~50°C, RS485 port, waterproofing connector, IP68.
Online multi-sensor flat head turbidity sensor	W510-C	Optical fiber turbidity, test range: 0.1~1000NTU, accuracy: <5% or 0.3NTU, response time: <2 sec, sheath material: 316L, operating temperature: :0~50°C, RS485 port, waterproofing connector, IP68.
Online multi-sensor digital pH sensor	W532-A	Online industrial pH sensor, pH 0-12, RS485 output, waterproofing connector, IP68.
Online multi-sensor automatic cleaning brush	W541-A	Automatic cleaning brush kit, including motor brush and positioning device, 2 sets of components, waterproofing connector, IP68.
Online multi-sensor body	W603-A	Online multi-sensor body, can connect pH, four-electrode conductivity, fluorescence dissolved oxygen, turbidity, including protective cover, RS485 port, waterproofing connector, IP68.

W4000 All-in-One Water Quality Sensor Structure Diagram



W4000 Technical Specification

Fluorescence Dissolved Oxygen Sensor	Test Range	0-20mg/L or 0-200% saturation	
	Accuracy	±0.3mg/L	
	Resolution	0.01mg/L	
Optical Fiber Turbidity Sensor	Test Range	0-1000 NTU	
	Accuracy	<5% or 0.3NTU	
	Resolution	0.1NTU	
Four-electrode Conductivity Sensor	Test Range	1uS/cm-200 mS/cm	1uS/cm-100 mS/cm
	Accuracy	1% FS	
	Resolution	0-1000uS/cm: 0.1uS/cm; 1mS/cm-100mS/cm: 0.1mS/cm	
Digital pH Sensor	Test Range	0-14pH	
	Accuracy	±0.1pH	
	Resolution	0.01	
Other Information of Multi-parameter Sensor	Operating Temperature	0~50°C	
	Output	One channel RS485 output	
	Automatic Cleaning Brush	Yes/optional	
	Power Supply	12V ±5%	
	Current	100 mA	
	Protection Grade	IP68	
	Cable Length	Standard 10m, 5m, 15m and 30m cable can be customized	

W4000 Maintenance Methods and Frequently Asked Questions

I. Maintenance schedule and methods

1. Maintenance Schedule

W4000 all-in-one water quality sensor with cleaning brush for extended maintenance intervals. However, due to the diversity of the environment, it is recommended to check, clean and calibrate the sensor regularly.

Maintenance Tasks	Recommended Maintenance Frequency
Cleaning Sensor	Depending on the usage environment
Calibration (if required)	Calibrate the sensor regularly

2. Maintenance Method

- 1) Check: Check whether there is dirt and microbial attachment on the head of the sensor, whether the outer casing and sensor surface are damaged, whether the cable is normal, whether the test data are normal, whether the consumables are damaged.
- 2) Clean: Wash the outer surface of the sensor with tap water, if there is still debris left, wipe it with a soft and damp cloth. For some stubborn dirt, you can add some household washing liquid to the tap water to clean.
- 3) Calibration: Single point or two points calibration for sensors. Select the appropriate standard solution according to the corresponding sensor.

II. Precautions

The sensor contains sensitive optical components and electronic components. Make sure the sensor is not subjected to severe mechanical shock. There are no parts inside the sensor that require maintenance.

III. FAQ

Error	Possible Cause	Solution
The operation interface cannot be connected or the measurement result is not displayed.	Controller and cable connection error	Reconnect controller and cable
	Cable failure	Please contact with us
The measured value is too high, too low or the value is continuously unstable	The sensor has dirt and microbial adhesion	Cleaning the sensor surface

W4000 Cable Definition

1. Sensor Dimensions
100x381 mm(Φ x L)
2. Power Supply Requirements
The power supply must be DC 12V $\pm$ 5%.
3. Sensor Cable
4-wire AWG-24 or AWG-26 shielding wire.





1. Red—Power (VCC)
2. White—485 Date_B (485_B)
3. Green ---485 Date_A (485_A)
4. Black --- Ground (GND)
5. Bare wire ---- Shield

Quality Assurance

Warranty Period	
Fluorescence Dissolved Oxygen Sensor	One Year
Flat Head Turbidity Sensor	One Year
Four-electrode Conductivity Sensor	One Year
Digital pH Sensor	Three Months
Temperature	One Year
Online Multi-sensor Body	Three Years
Other Consumables	Three Months

Yantai Winmore guarantees to its first-hand purchaser that there are no product defects caused by material failure or factory manufacturing during the validity period from the date of shipment.

If there are defects found during the warranty period, Yantai Winmore promises to repair or replace the defective products, or return the payment of product except the charge for the first time for transport and related formalities. In the warranty period, repair or replacement of any product will only enjoy the rest of the original warranty.

This warranty does not apply to consumables, such as the consumption parts (including but not limited to the lamp, piping, etc.). Contact Winmore or your agents to start technical support within the guarantee period.

After receiving feedback for the product quality problems from the customer, Yantai Winmore will confirm whether the product need repair within two weeks; it can't be returned without approval to repair the product.

Limitation of Warranty

This warranty does not include the following:

- Damage caused due to force majeure, natural disasters, social unrest, war(published or unpublished), terrorism, civil war or any government forced.
- Damage caused due to improper use, negligence, accident, or caused by the improper application and installation.
- Freight for the product shipped back to Winmore.
- Freight for parts or products express or express delivery within the warranty.
- Travel expense for repair in local in warranty.

The quality assurance includes all content of products provided by Winmore.

It constitutes the final, complete and exclusive statement about the quality guarantee, no person or agent is authorized in the name of Winmore to develop other warranty.

As described above, the remedial measures such as repair, replacement or return the payment for product is not in violation of the warranty, and it aim at our own products only. Based on the strict liability or other legal theory, Winmore is not responsible for defects or any other damage due to careless operation , including the subsequent damage with a causal connection between these situations.